Cavitation Bubble Collapse: Rayleigh–Plesset Dynamics with an Adaptive Runge–Kutta Integrator

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1 Premise

1.1 Motivation: bubbles in rocket turbopumps

A liquid rocket engine needs its propellants delivered at high pressure, and the turbopump that does this is one of the hardest parts of the engine to design. To keep the propellant tanks light, the tank pressure is kept low, which means the liquid arrives at the pump with only a small margin above its vapor pressure. The first stage of the pump, the *inducer*, is an axial impeller built to run with some vapor present. Near its blade leading edges and tip clearance the local pressure drops below the vapor pressure, and small gas and vapor nuclei in the liquid grow into cavitation bubbles [1, 2].

Those bubbles are then carried downstream into higher pressure, and they collapse. A collapsing bubble focuses the energy of the surrounding liquid into a tiny volume over a very short time, producing high wall speeds and large local pressures. Repeated near a surface, this pits and erodes blades. At the pump level, enough cavitation degrades the head the pump delivers and can drive unsteady behavior such as rotating cavitation and cavitation surge [1]. The same physics shows up in propellant feed lines, valves and injectors.

Real inducer cavitation involves clouds and sheets of bubbles, walls and non-spherical collapse. We study the simplest building block: a single spherical bubble in a large body of liquid whose far-field pressure changes in time. The Rayleigh–Plesset equation describes this bubble [3–5]. It is simple enough to study carefully and still captures the violent, strongly nonlinear collapse that makes cavitation damaging. We use water at 20 °C as the working liquid because its properties are well known and the literature for checking our results is largest. Cryogenic propellants behave differently, partly because evaporation and condensation cool and heat the bubble (the "thermal effect" that suppresses cavitation in cryogens [5]), and our model leaves that out.

1.2 Research questions

Q1. How hard does a bubble collapse after a pressure jump? A nucleus that has settled to equilibrium in the low-pressure region near a blade is suddenly exposed to a higher recovery pressure. How do the minimum radius $R_{\min}/R_0$, the peak inward wall speed $|\dot{R}|_{\max}$ and the peak gas pressure $p_{g,\max}$ scale with the pressure ratio $\Pi = p_1/p_0$ and the initial radius R_0, and where does the predicted collapse become too fast for the incompressible model?

Q2. How does a bubble respond to oscillating pressure, and where does it stop behaving linearly? With the ambient pressure oscillating at frequency ω and relative amplitude ε, how does the steady response near the small-amplitude natural frequency ω_0 change as ε grows? Does the frequency of largest response move away from ω_0, and does the oscillation become asymmetric, with larger growth than compression?

Q3. Which physics controls the collapse, and when? Gas compression, viscosity and surface tension all resist collapse. How much do viscosity and surface tension each change $R_{\min}$, and how does that depend on the bubble size R_0? We answer this by switching each term off in turn and comparing against the dimensionless groups of Section 1.5.

Q1 and Q2 are the core of the project; Q3 is a smaller study built on the Case B setup of Q1. None of these questions can be answered from a closed-form solution, as explained in Section 1.6.

1.3 The initial value problem

The bubble radius $R(t)$ obeys the Rayleigh–Plesset equation [4–6]

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho}\left[p_v + p_{g0}\left(\frac{R_0}{R}\right)^{3\kappa} - p_\infty(t) - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R}\right], \qquad R(0) = R_0, \quad \dot{R}(0) = 0, \tag{1}$$

where dots are time derivatives. The bracket is the pressure in the liquid just outside the bubble wall minus the far-field pressure. The initial gas pressure is set so the bubble starts in equilibrium with the ambient pressure at $t = 0$,

$$p_{g0} = p_\infty(0) - p_v + \frac{2\sigma}{R_0}, \tag{2}$$

except in Case A, where the bubble is empty ($p_{g0} = 0$). With the state $\mathbf{y} = [R,\ \dot{R}]^\mathsf{T}$, Eq. (1) becomes the first-order system

$$\frac{d}{dt}\begin{bmatrix} R \\ \dot{R} \end{bmatrix} = \begin{bmatrix} \dot{R} \\ \dfrac{1}{R}\left\{\dfrac{1}{\rho}\left[p_v + p_{g0}\left(\dfrac{R_0}{R}\right)^{3\kappa} - p_\infty(t) - \dfrac{2\sigma}{R} - \dfrac{4\mu\dot{R}}{R}\right] - \dfrac{3}{2}\dot{R}^2\right\} \end{bmatrix}, \tag{3}$$

which is what our integrators advance.

Symbols. R is the bubble radius and R_0 its initial value; ρ, μ and σ are the liquid density, dynamic viscosity and surface tension; p_v is the vapor pressure, taken constant; p_{g0} is the partial pressure of non-condensable gas when $R = R_0$; κ is the polytropic exponent of that gas ($\kappa = 1$ isothermal, $\kappa = 1.4$ adiabatic air); and $p_\infty(t)$ is the prescribed far-field liquid pressure. Table 1 lists values.

Units check. Multiplying Eq. (1) by ρ, every term should be a pressure. On the left, $\rho R\ddot{R}$ and $\rho\dot{R}^2$ have units $(\mathrm{kg/m^3})(\mathrm{m^2/s^2}) = \mathrm{Pa}$. On the right, p_v, p_{g0} and p_∞ are pressures (the factor $(R_0/R)^{3\kappa}$ is dimensionless), $2\sigma/R$ is $(\mathrm{N/m})/\mathrm{m} = \mathrm{Pa}$, and $4\mu\dot{R}/R$ is $(\mathrm{Pa\,s})(\mathrm{m/s})/\mathrm{m} = \mathrm{Pa}$. All terms are consistent.

Table 1: Parameters. Liquid properties are for water at 20 °C [8]. "Fixed" parameters are held at the baseline except when a term is switched off for Q3.

Symbol	Meaning	Units	Baseline	Range swept
ρ	liquid density	kg/m^3	998.2	fixed
μ	dynamic viscosity	Pa s	1.0016×10^{-3}	fixed, or 0
σ	surface tension	N/m	0.07274	fixed, or 0
p_v	vapor pressure	Pa	2.339×10^3	fixed
p_0	ambient pressure before the event	Pa	1.01325×10^5	fixed
κ	polytropic exponent	–	1.4	fixed
R_0	initial radius	m	100 μm (B), 10 μm (C)	1 μm to 1 mm
$\Pi = p_1/p_0$	pressure ratio, Case B	–	5	2 to 20
ε	forcing amplitude, Case C	–	0.01	10^{-4} to 0.3
ω/ω_0	forcing frequency, Case C	–	–	0.5 to 2

Assumptions. The bubble stays spherical; the liquid is incompressible and infinite in extent; the gas compresses polytropically with fixed κ, with no gas diffusion across the wall; the vapor stays at p_v (evaporation and condensation are instantaneous); and the gas pressure is uniform inside the bubble. Incompressibility is the assumption most likely to fail. Equation (1) is the zero-Mach-number limit of the compressible bubble equations (Keller–Miksis, for example, adds corrections of first order in the wall Mach number $|\dot{R}|/c$) [7], so its error grows as the wall speed approaches the sound speed $c \approx 1482$ m/s in water [8]. There is no single accepted cutoff, so we set our own: we report $|\dot{R}|_{\max}/c$ for every run and treat results with $|\dot{R}|_{\max}/c > 0.1$ as outside the model's range.

1.4 Test cases

A bubble that starts in equilibrium under constant ambient pressure never moves, so each case below is built to produce real dynamics: an empty cavity, a pressure step, or oscillating forcing.

Case A: empty-cavity (Rayleigh) collapse. Set $p_{g0} = 0$, $p_v = 0$, $\sigma = \mu = 0$ and $p_\infty = \Delta p$ constant. The bubble starts at rest with nothing inside it, so it collapses. Multiplying Eq. (1) by $2R^2\dot{R}$ and integrating once gives

$$\dot{R}^2 = \frac{2\Delta p}{3\rho}\left(\frac{R_0^3}{R^3} - 1\right), \tag{4}$$

and integrating again gives the exact time to reach any radius R,

$$t(R) = R_0\sqrt{\frac{3\rho}{2\Delta p}}\int_{R/R_0}^{1} \frac{x^{3/2}}{\sqrt{1-x^3}}\,dx. \tag{5}$$

Setting $R = 0$ and substituting $u = x^3$ turns the integral into a Beta function, so the collapse time is

$$t_c = \frac{1}{3}\sqrt{\frac{3}{2}}\,B\left(\tfrac{5}{6}, \tfrac{1}{2}\right) R_0\sqrt{\frac{\rho}{\Delta p}} \approx 0.91468\, R_0\sqrt{\frac{\rho}{\Delta p}}, \tag{6}$$

which matches the constant 0.915 given by Rayleigh [3] and Brennen [5]. For $R_0 = 1$ mm and $\Delta p = 101.325$ kPa, $R_0\sqrt{\rho/\Delta p} = 99.3$ μs and $t_c \approx 90.8$ μs. Equation (5) gives a reference value of R at any time, not just t_c, so it checks the whole trajectory. This case is our main correctness check. It is also a hard test for a time integrator, because the solution is singular at t_c ($\dot{R} \to -\infty$ as $R \to 0$), so we compare up to a small final radius rather than all the way to t_c.

Case B: gas bubble after a pressure step. The bubble sits in equilibrium at $p_\infty = p_0$, so p_{g0} follows from Eq. (2) with p_0. At $t = 0$ the far-field pressure jumps to $p_1 = \Pi p_0$ and stays there:

$$p_\infty(t) = p_1 \quad \text{for } t > 0. \tag{7}$$

This mimics a nucleus convected out of the low-pressure zone near a blade into the recovery region. At $t = 0^+$ the wall acceleration is $\ddot{R} = (p_0 - p_1)/(\rho R_0) < 0$, so the bubble starts collapsing at once. The gas inside stiffens rapidly as R drops, stops the collapse at a finite $R_{\min}$, and the bubble rebounds and oscillates. The right-hand side is smooth for $t > 0$, which makes this a good case for convergence studies. With $\mu = 0$, there is an exact conservation law to check against: the energy

$$E = 2\pi\rho R^3\dot{R}^2 + \frac{4\pi}{3}(p_1 - p_v)R^3 + 4\pi\sigma R^2 + \frac{4\pi}{3}\,\frac{p_{g0}R_0^{3\kappa}R^{3(1-\kappa)}}{\kappa - 1} \tag{8}$$

(liquid kinetic energy, work against the far field, surface energy and gas internal energy) is constant in time for $\kappa \neq 1$. Differentiating E and using Eq. (1) confirms $dE/dt = 0$. Case B is also the workhorse for Q1 and Q3.

Case C: oscillating ambient pressure. The bubble starts in equilibrium at p_0, and the far-field pressure oscillates:

$$p_\infty(t) = p_0\,[1 - \varepsilon\sin(\omega t)]\,. \tag{9}$$

Linearizing Eq. (1) about R_0 with $R = R_0(1 + x)$, $|x| \ll 1$, gives a damped, driven oscillator

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = \frac{\varepsilon p_0}{\rho R_0^2}\sin(\omega t), \qquad \omega_0^2 = \frac{1}{\rho R_0^2}\left(3\kappa p_{g0} - \frac{2\sigma}{R_0}\right), \qquad \beta = \frac{2\mu}{\rho R_0^2}. \tag{10}$$

With $\sigma = p_v = 0$ this reduces to the Minnaert frequency $\omega_0 = (1/R_0)\sqrt{3\kappa p_0/\rho}$ [9], about $f_0 R_0 \approx 3.29$ m/s for air in water at one atmosphere. For our baseline $R_0 = 10$ μm with all terms kept, $f_0 = \omega_0/2\pi \approx 342$ kHz for $\kappa = 1.4$ (288 kHz for $\kappa = 1$), and the quality factor is $\omega_0/(2\beta) \approx 54$. The steady linear amplitude, $|x| = (\varepsilon p_0/\rho R_0^2)/\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$, is an exact reference at small ε. Increasing ε takes us into the nonlinear regime, where the response is no longer symmetric and the peak can shift, which is Q2.

1.5 Dimensionless form

Scale the radius by R_0 and time by the Rayleigh time t_R,

$$\tilde{R} = \frac{R}{R_0}, \qquad \tilde{t} = \frac{t}{t_R}, \qquad t_R = R_0\sqrt{\frac{\rho}{\Delta p}}, \tag{11}$$

where Δp is the pressure difference driving the motion: $\Delta p = p_1 - p_v$ for Case B ($\Delta p = p_\infty$ for Case A, where $p_v = 0$), and $\Delta p = p_0 - p_v$ for Case C. With primes for $d/d\tilde{t}$, Eq. (1) becomes

$$\tilde{R}\tilde{R}'' + \frac{3}{2}\tilde{R}'^2 = P_g\,\tilde{R}^{-3\kappa} - P(\tilde{t}) - \frac{S}{\tilde{R}} - \frac{4}{\mathrm{Re}}\,\frac{\tilde{R}'}{\tilde{R}}, \qquad \tilde{R}(0) = 1, \quad \tilde{R}'(0) = 0, \tag{12}$$

with the dimensionless groups

$$P_g = \frac{p_{g0}}{\Delta p}, \qquad P(\tilde{t}) = \frac{p_\infty(t) - p_v}{\Delta p}, \qquad S = \frac{2\sigma}{R_0\,\Delta p}, \qquad \mathrm{Re} = \frac{R_0\sqrt{\rho\,\Delta p}}{\mu}, \tag{13}$$

plus κ, and for Case C the forcing amplitude ε and frequency ωt_R (or equivalently ω/ω_0, since $(\omega_0 t_R)^2 = 3\kappa P_g - S$). In Case A, $P_g = S = 1/\mathrm{Re} = 0$ and $P = 1$, so the dimensionless collapse time is the pure number 0.91468. In Case B, $P = 1$ and $P_g = (p_0 - p_v)/(p_1 - p_v) + S \approx 1/\Pi$, so the pressure ratio, κ, S and Re fully determine $\tilde{R}(\tilde{t})$.

For water at one atmosphere, $S = 1$ at $R_0 \approx 1.5$ μm and $\mathrm{Re} = 1$ at $R_0 \approx 0.1$ μm. For $R_0 \gtrsim 10$ μm both corrections are small at the scale of the initial bubble ($S \approx 0.15$, $\mathrm{Re} \approx 100$ at 10 μm), so Case B results for different bubble sizes should fall onto one curve when plotted against Π in these variables, and departures from that curve at small R_0 show where surface tension and viscosity take over. One caution for Q3: these groups use the initial radius and the bulk pressure. Near $R_{\min}$ the local length scale is much smaller and the local speed much larger, so viscosity and surface tension can matter more at the instant of collapse than S and Re suggest.

1.6 Why we need numerical methods

Equation (1) is a nonlinear second-order ODE with no closed-form solution once gas pressure, surface tension or viscosity are included, or once the ambient pressure varies in time. The only exact solution is the Case A limit, and even there $R(t)$ is known only implicitly through Eq. (5). The linearization in Eq. (10) holds only for tiny oscillations, which is exactly the regime our questions are not about: Q1 asks for $R_{\min}$ and peak pressures in collapses where R changes by a large factor, and Q2 asks where the linear prediction stops working. So we need to integrate the IVP numerically, across many parameter combinations.

The problem is also hard for a time integrator in a specific way. Near the end of a collapse, Eq. (4) gives $|\dot{R}| \propto R^{-3/2}$, so $R \propto (t_c - t)^{2/5}$, and the local time scale $R/|\dot{R}| = \frac{5}{2}(t_c - t)$ shrinks to zero. In a gas-filled bubble the collapse is stopped at $R_{\min}$, but the time spent near the minimum is still a small fraction of the overall period, and it gets smaller as Π grows. A fixed step small enough to resolve the minimum wastes most of its work on the slow parts of the motion, which is why we compare fixed-step RK4 with an adaptive embedded Runge–Kutta pair that can take small steps only where they are needed.

2 Method

We advance the initial value problem of Section 1.3 with explicit Runge–Kutta (RK) methods that we implemented ourselves. We use two of them: the classical fixed-step fourth-order method (RK4) and the adaptive Dormand–Prince 5(4) embedded pair (DP5). This section explains why that choice fits our questions, derives the methods and their error and stability properties, and gives the algorithm that takes a step from t_k to t_{k+1}.

2.1 The problem in the form the integrator sees

It is convenient to integrate the dimensionless system of Eq. (12), with state $\mathbf{y} = [\tilde{R}, \tilde{R}']^T$ and $\tilde{t}$ as the independent variable. Every component is then $O(1)$ over most of the motion, so one tolerance treats both components fairly (in SI units $R \sim 10^{-5}\,\mathrm{m}$ and $\dot{R} \sim 10^{2}\,\mathrm{m/s}$ would need separate scalings). Writing $\mathbf{y}' = \mathbf{f}(\tilde{t}, \mathbf{y})$,

$$\mathbf{f}(\tilde{t}, \mathbf{y}) = \begin{bmatrix} y_2 \\ \dfrac{1}{y_1}\left\{ P_g\, y_1^{-3\kappa} - P(\tilde{t}) - \dfrac{S}{y_1} - \dfrac{4}{\mathrm{Re}}\dfrac{y_2}{y_1} - \dfrac{3}{2}\, y_2^2 \right\} \end{bmatrix}, \qquad \mathbf{y}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{14}$$

For $\tilde{t} > 0$ the right-hand side is smooth wherever $y_1 > 0$. In Case B the pressure step happens exactly at $\tilde{t} = 0$ and P is constant afterwards, so the integrator never crosses a discontinuity. In Case C the forcing $P(\tilde{t})$ is a smooth sinusoid. The only singular behavior is $y_1 \to 0$, which we discuss in Section 2.7.

2.2 Why an explicit Runge–Kutta method

We judge the method on the three criteria that our questions impose.

Accuracy. Q1 asks for $R_{\min}/R_0$, $|\dot{R}|_{\max}$ and $p_{g,\max}$. These are extreme values reached in the last fraction of the collapse, where the solution changes fastest. They are also amplified by the physics: since $p_g \propto R^{-3\kappa}$, a relative error δ in $R_{\min}$ produces a relative error of about $3\kappa\,\delta$ in $p_{g,\max}$ (about 4δ for $\kappa = 1.4$). Q2 needs the steady response after many forcing periods, which requires phase errors to stay small over a long run. A high-order method is the efficient way to meet both demands, because the error falls as h^p with p large.

Stability. Explicit methods are only efficient if the problem is not stiff. Linearizing about equilibrium (Eq. (10)) gives eigenvalues $\lambda = -\beta \pm i\sqrt{\omega_0^2 - \beta^2}$ with $\beta/\omega_0 \lesssim 0.1$ for $R_0 \gtrsim 1\,\mu$m in water. The spectrum therefore lies close to the imaginary axis with a modest real part, and there is no fast decaying mode that would force a tiny step for stability alone. The stability limit $h|\lambda| \lesssim 2.8$ is far looser than the step needed to resolve the collapse, so the step size is set by accuracy, not stability. An implicit method would cost a nonlinear solve per step and buy us nothing here. The linearization is only valid near equilibrium, so we also check this along real collapse trajectories in Section 3.3: there $|h\lambda|$ stays below 0.17 at every accepted step.

Cost and a shrinking time scale. We need many runs across (R_0, Π) and (ε, ω). Near the end of a collapse the local time scale $\tilde{R}/|\tilde{R}'| = \frac{5}{2}(\tilde{t}_c - \tilde{t})$ (Section 1.6) shrinks toward zero, while the rest of the motion is slow. A fixed step small enough for the minimum wastes most of its steps on the slow part. An adaptive method takes small steps only where the local error requires them. One-step methods such as RK are the natural choice for this because the step size can change freely at every step. Linear multistep methods (Adams–Bashforth–Moulton) need equally spaced history and are awkward to restart each time the step changes, which would happen constantly in a collapse.

Why two methods. Fixed-step RK4 is simple and its order of accuracy is easy to demonstrate, so we use it for the convergence study. DP5 is the production integrator for the parameter sweeps. Comparing their cost at equal accuracy is a result in its own right.

2.3 Derivation of Runge–Kutta methods

Let $\mathbf{y}' = \mathbf{f}(t, \mathbf{y})$ and let $\mathbf{y}_k \approx \mathbf{y}(t_k)$ with step $h = t_{k+1} - t_k$. Integrating the ODE over one step gives the exact identity

$$\mathbf{y}(t_{k+1}) = \mathbf{y}(t_k) + \int_{t_k}^{t_{k+1}} \mathbf{f}(t, \mathbf{y}(t))\, dt. \tag{15}$$

A Runge–Kutta method replaces the integral by a quadrature rule with weights b_i at nodes $t_k + c_i h$, and estimates the unknown values of $\mathbf{y}$ at those nodes by earlier stages. An s-stage explicit method

is

$$\mathbf{k}_i = \mathbf{f}\Big(t_k + c_i h,\ \mathbf{y}_k + h\sum_{j=1}^{i-1} a_{ij}\mathbf{k}_j\Big), \qquad i = 1, \dots, s, \tag{16}$$

$$\mathbf{y}_{k+1} = \mathbf{y}_k + h\sum_{i=1}^{s} b_i \mathbf{k}_i, \tag{17}$$

where $c_i = \sum_j a_{ij}$. The coefficients are collected in a Butcher tableau. They are chosen so that the Taylor expansion of $\mathbf{y}_{k+1}$ in powers of h agrees with the Taylor expansion of the exact solution $\mathbf{y}(t_{k+1})$ through as many orders as possible. Matching the terms up to h^p gives the order conditions. For $p \le 4$ they are, with the sums running over all stage indices,

$$\begin{aligned} &\textstyle\sum_i b_i = 1, && \textstyle\sum_i b_i c_i = \tfrac{1}{2}, \\ &\textstyle\sum_i b_i c_i^2 = \tfrac{1}{3}, && \textstyle\sum_{i,j} b_i a_{ij} c_j = \tfrac{1}{6}, \\ &\textstyle\sum_i b_i c_i^3 = \tfrac{1}{4}, && \textstyle\sum_{i,j} b_i c_i a_{ij} c_j = \tfrac{1}{8}, \\ &\textstyle\sum_{i,j} b_i a_{ij} c_j^2 = \tfrac{1}{12}, && \textstyle\sum_{i,j,l} b_i a_{ij} a_{jl} c_l = \tfrac{1}{24}. \end{aligned} \tag{18}$$

The first condition gives order 1 (consistency), the first two give order 2, the first four give order 3, and all eight give order 4 [10, 11].

Classical RK4. The choice $c = (0, \frac{1}{2}, \frac{1}{2}, 1)$, $a_{21} = a_{32} = \frac{1}{2}$, $a_{43} = 1$ and $b = (\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6})$ satisfies all eight conditions in Eq. (18). For example,

$$\textstyle\sum_i b_i c_i = \frac{1}{3}\cdot\frac{1}{2} + \frac{1}{3}\cdot\frac{1}{2} + \frac{1}{6}\cdot 1 = \frac{1}{2}, \qquad \sum_{i,j,l} b_i a_{ij} a_{jl} c_l = b_4 a_{43} a_{32} c_2 = \frac{1}{6}\cdot 1\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{24}.$$

Written out, one step of RK4 is

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(t_k, \mathbf{y}_k), & \mathbf{k}_2 &= \mathbf{f}(t_k + \tfrac{h}{2}, \mathbf{y}_k + \tfrac{h}{2}\mathbf{k}_1), \\ \mathbf{k}_3 &= \mathbf{f}(t_k + \tfrac{h}{2}, \mathbf{y}_k + \tfrac{h}{2}\mathbf{k}_2), & \mathbf{k}_4 &= \mathbf{f}(t_k + h, \mathbf{y}_k + h\mathbf{k}_3), \\ & & \mathbf{y}_{k+1} &= \mathbf{y}_k + \tfrac{h}{6}\,(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned} \tag{19}$$

Embedded pairs and Dormand–Prince 5(4). An embedded pair uses one set of stages $\mathbf{k}_i$ and two sets of weights, b_i of order p and b_i^* of order $p-1$:

$$\mathbf{y}_{k+1} = \mathbf{y}_k + h\sum_i b_i \mathbf{k}_i, \qquad \mathbf{y}^*_{k+1} = \mathbf{y}_k + h\sum_i b_i^* \mathbf{k}_i. \tag{20}$$

The two results come from the same stages, so the second costs almost nothing. DP5 [12] has $s = 7$ stages with $p = 5$, and its coefficients are in Table 2. It has the first-same-as-last (FSAL) property: the last stage $\mathbf{k}_7 = \mathbf{f}(t_{k+1}, \mathbf{y}_{k+1})$ equals $\mathbf{k}_1$ of the next step, so an accepted step costs six new evaluations of $\mathbf{f}$, not seven.

2.4 Error properties

Subtracting the Taylor expansions of the numerical and exact solutions, a method of order p has a local truncation error (error made in one step starting from the exact solution)

$$\boldsymbol{\tau}_{k+1} = \mathbf{y}(t_{k+1}) - \hat{\mathbf{y}}_{k+1} = C(t_k)\, h^{p+1} + O(h^{p+2}), \tag{21}$$

0							
$\frac{1}{5}$	$\frac{1}{5}$						
$\frac{3}{10}$	$\frac{3}{40}$	$\frac{9}{40}$					
$\frac{4}{5}$	$\frac{44}{45}$	$-\frac{56}{15}$	$\frac{32}{9}$				
$\frac{8}{9}$	$\frac{19372}{6561}$	$-\frac{25360}{2187}$	$\frac{64448}{6561}$	$-\frac{212}{729}$			
1	$\frac{9017}{3168}$	$-\frac{355}{33}$	$\frac{46732}{5247}$	$\frac{49}{176}$	$-\frac{5103}{18656}$		
1	$\frac{35}{384}$	0	$\frac{500}{1113}$	$\frac{125}{192}$	$-\frac{2187}{6784}$	$\frac{11}{84}$	
b_i (order 5)	$\frac{35}{384}$	0	$\frac{500}{1113}$	$\frac{125}{192}$	$-\frac{2187}{6784}$	$\frac{11}{84}$	0
b_i^* (order 4)	$\frac{5179}{57600}$	0	$\frac{7571}{16695}$	$\frac{393}{640}$	$-\frac{92097}{339200}$	$\frac{187}{2100}$	$\frac{1}{40}$

Table 2: Butcher tableau of the Dormand–Prince 5(4) pair. The first column holds the nodes c_i and the entries to its right are a_{ij}; the last two rows are the weights.

where C depends on the elementary differentials of $\mathbf{f}$ at the order-$(p+1)$ trees, which the method's coefficients fail to match. Over a fixed interval $[0, T]$ there are T/h steps. If $\mathbf{f}$ is Lipschitz with constant L and sufficiently smooth on the region the trajectory visits, the local errors accumulate (each amplified by at most e^{LT}) into a global error

$$\|\mathbf{y}(t_k) - \mathbf{y}_k\| \le \frac{C_{\max}}{L}\left(e^{LT} - 1\right) h^p = O(h^p). \tag{22}$$

So RK4 is globally fourth order ($p = 4$) and DP5 is fifth order ($p = 5$). This is the slope we test in Section 3. Two caveats matter here. First, the constants C and L grow without bound as $\tilde{R} \to 0$, because $\mathbf{f}$ is not Lipschitz there. The h^p rate holds only once h is small enough to resolve the minimum, and the convergence study needs to be done in that regime. Second, the factor e^{LT} makes the bound very pessimistic for long runs. For the many-period oscillations of Case C the error in practice accumulates mainly as phase error, which grows roughly linearly with the number of periods, so it is the phase error that sets h there.

Error estimate for the adaptive pair. Since $\mathbf{y}_{k+1}$ is fifth order and $\mathbf{y}^*_{k+1}$ is fourth order,

$$\mathbf{y}^*_{k+1} - \mathbf{y}_{k+1} = (\mathbf{y}^*_{k+1} - \hat{\mathbf{y}}_{k+1}) - (\mathbf{y}_{k+1} - \hat{\mathbf{y}}_{k+1}) = O(h^5), \tag{23}$$

dominated by the local error of the fourth-order solution. The difference

$$\mathbf{e}_{k+1} = h \sum_{i=1}^{7} (b_i - b_i^*)\mathbf{k}_i \tag{24}$$

is therefore a computable estimate of the local error, with no extra evaluations of $\mathbf{f}$. We then advance with the more accurate fifth-order value $\mathbf{y}_{k+1}$ (local extrapolation). This makes the estimate slightly conservative.

2.5 Stability properties

Applying an explicit RK method to the test equation $y' = \lambda y$ gives $y_{k+1} = R(z)\, y_k$ with $z = h\lambda$, where R is a polynomial. The method is absolutely stable when $|R(z)| \le 1$. For RK4 ($s = p = 4$),

$$R_{\mathrm{RK4}}(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}, \tag{25}$$

and for DP5

$$R_{\mathrm{DP5}}(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} + \frac{z^5}{120} + \frac{z^6}{600}. \tag{26}$$

The stability interval on the negative real axis is about $z \in [-2.78, 0]$ for RK4 and about $[-3.3, 0]$ for DP5. The imaginary axis matters for us because the linearized bubble oscillator has eigenvalues near it (Section 2.2). RK4 contains the imaginary segment $|\mathrm{Im}\, z| \le 2\sqrt{2} \approx 2.83$. Because $|R_{\mathrm{RK4}}(i\theta)|^2 = 1 - \theta^6/72 + \theta^8/576 < 1$ for small $\theta \neq 0$, RK4 adds a small numerical damping of about $\theta^6/144$ per step to an undamped oscillation, which is $O(h^5)$ per unit time; its phase error, $\theta^5/120$ per step, is $O(h^4)$ per unit time and is the larger of the two. DP5 behaves differently: $|R_{\mathrm{DP5}}(iy)| \le 1$ only for $|y| \lesssim 0.998$, and for larger $|y|$ it slowly amplifies an undamped oscillation. So for DP5 the binding limit on oscillatory modes is $|h\lambda| < 1$, not the real-axis value 3.3. In our runs the accuracy-limited step keeps $|h\lambda|$ several times below this (Section 3.3), so accuracy, not stability, is the binding constraint for both methods.

2.6 Adaptive step-size control

Let atol and rtol be absolute and relative tolerances. For each component i of the state, define the scale

$$\mathrm{sc}_i = \mathrm{atol} + \mathrm{rtol} \cdot \max(|y_{k,i}|,\, |y_{k+1,i}|), \tag{27}$$

and the scaled error norm

$$E = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_{k+1,i}}{\mathrm{sc}_i}\right)^2}, \qquad n = 2. \tag{28}$$

The step is accepted if $E \le 1$ and rejected otherwise. Because the local error behaves like $E \propto h^5$, the step size that would give $E = 1$ is $h\,E^{-1/5}$. The next step is therefore

$$h_{\mathrm{new}} = h \cdot \min\Big(f_{\max},\, \max(f_{\min},\, \gamma\, E^{-1/5})\Big), \tag{29}$$

with a safety factor $\gamma = 0.9$, and limits $f_{\min} = 0.2$, $f_{\max} = 5$ that stop the step from changing too abruptly. After a rejected step the factor is also capped at 1. We control the *local* error per step, not the global error, so the global error is typically a modest multiple of the tolerance and can grow in long runs; we measure it against the exact and conservation checks of Section 3 and do not assume it equals the tolerance. The first step is chosen with the standard starting-step estimate of [10] (one extra evaluation of $\mathbf{f}$), and the last step is shortened to land exactly on the final time. Based on Section 3.4 we use $\mathrm{rtol} = 10^{-8}$ and $\mathrm{atol} = 10^{-10}$ for all production runs.

2.7 Handling the collapse singularity

As $\tilde{R} \to 0$ the right-hand side of Eq. (14) blows up. For a gas bubble the gas pressure stops the collapse at a finite $\tilde{R}_{\min} > 0$, so a solution exists and the adaptive controller shrinks h automatically as the local time scale shrinks. We add three safeguards. (i) A step that produces $y_1 \le 0$ or a non-finite value is rejected and retried with a quarter of the step. (ii) The step is bounded below by $h_{\min}$, and the run is flagged as unreliable if $h_{\min}$ is reached. (iii) In Case A (empty cavity) the true solution is singular at $\tilde{t}_c$, so we stop the integration when $\tilde{R}$ first falls below a chosen final radius and compare there, as described in Section 1.4. To locate $\tilde{R}_{\min}$ and $|\tilde{R}'|_{\max}$ more accurately than the grid allows, we treat them as events: $\tilde{R}_{\min}$ is where $\tilde{R}'$ crosses zero from below and $|\tilde{R}'|_{\max}$ is where $\tilde{R}''$ does. When an accepted step brackets a sign change, we find the root by the Illinois (modified regula falsi) method, evaluating each trial point by re-taking a DP5 step of the trial length from

$\tilde{t}_k$. Every trial value is then a genuine fifth-order solution, so the located extrema are as accurate as the steps themselves, which a cubic Hermite interpolant ($O(h^4)$) would not guarantee. Case A's stopping radius is found the same way. We do not use a library integrator at any point.

2.8 Algorithm

Algorithm 1 advances one fixed step, and Algorithm 2 advances one adaptive step from $\tilde{t}_k$ to $\tilde{t}_{k+1}$. Here $\mathbf{f}$ is Eq. (14).

Algorithm 1 One step of RK4: $(t_k, \mathbf{y}_k, h) \mapsto (t_{k+1}, \mathbf{y}_{k+1})$

$\mathbf{k}_1 \leftarrow \mathbf{f}(t_k, \mathbf{y}_k)$
$\mathbf{k}_2 \leftarrow \mathbf{f}(t_k + h/2,\ \mathbf{y}_k + \frac{h}{2}\mathbf{k}_1)$
$\mathbf{k}_3 \leftarrow \mathbf{f}(t_k + h/2,\ \mathbf{y}_k + \frac{h}{2}\mathbf{k}_2)$
$\mathbf{k}_4 \leftarrow \mathbf{f}(t_k + h,\ \mathbf{y}_k + h\mathbf{k}_3)$
$\mathbf{y}_{k+1} \leftarrow \mathbf{y}_k + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4); \quad t_{k+1} \leftarrow t_k + h$

Algorithm 2 One adaptive DP5 step from t_k (with $\mathbf{k}_1 = \mathbf{f}(t_k, \mathbf{y}_k)$ carried over by FSAL)

repeat
  $h \leftarrow \min(h,\ t_{\text{end}} - t_k)$
  compute $\mathbf{k}_2, \ldots, \mathbf{k}_7$ from Eq. (16) using Table 2
  $\mathbf{y}_{k+1} \leftarrow \mathbf{y}_k + h\sum_i b_i \mathbf{k}_i$ (fifth order)
  $\mathbf{e} \leftarrow h\sum_i (b_i - b_i^*)\mathbf{k}_i$; compute E from the scaled norm
  $h_{\text{new}} \leftarrow h \cdot \min\big(f_{\max}, \max(f_{\min}, \gamma E^{-1/5})\big)$
  if $E \le 1$ **and** $y_{k+1,1} > 0$ **then**
    accept: $t_{k+1} \leftarrow t_k + h; \quad \mathbf{k}_1 \leftarrow \mathbf{k}_7; \quad h \leftarrow h_{\text{new}}$
  else
    reject: $h \leftarrow \min(h_{\text{new}},\ h)$, or $h/4$ if $y_{k+1,1} \le 0$ (if $h < h_{\min}$, flag and stop)
  end if
until step accepted

Cost. RK4 costs four evaluations of $\mathbf{f}$ per step and DP5 six per accepted step (plus six for each rejected step). Counting evaluations of $\mathbf{f}$ is a machine-independent measure of cost, and our integrators record it for every run so that fixed-step RK4 and adaptive DP5 can be compared at equal accuracy.

3 Correct implementation

We wrote both integrators ourselves in Python with NumPy; no library time integrator is called anywhere. The code separates the method from the problem: `src/rp/integrators.py` holds RK4, DP5 with a fixed step, and adaptive DP5 (Algorithms 1 and 2), each taking a generic right-hand side $\mathbf{f}(t, \mathbf{y})$ and counting its evaluations; `src/rp/model.py` holds Eq. (14), its Jacobian, the three test cases and the reference solutions; and every figure below is produced by one script in `experiments/`. A suite of 24 unit tests (`pytest`) checks the Butcher tableaus against the order conditions of Eq. (18) (and the nine fifth-order conditions for DP5), the stability polynomials, the nondimensionalization against the SI form of Eq. (1), and each verification result reported here.

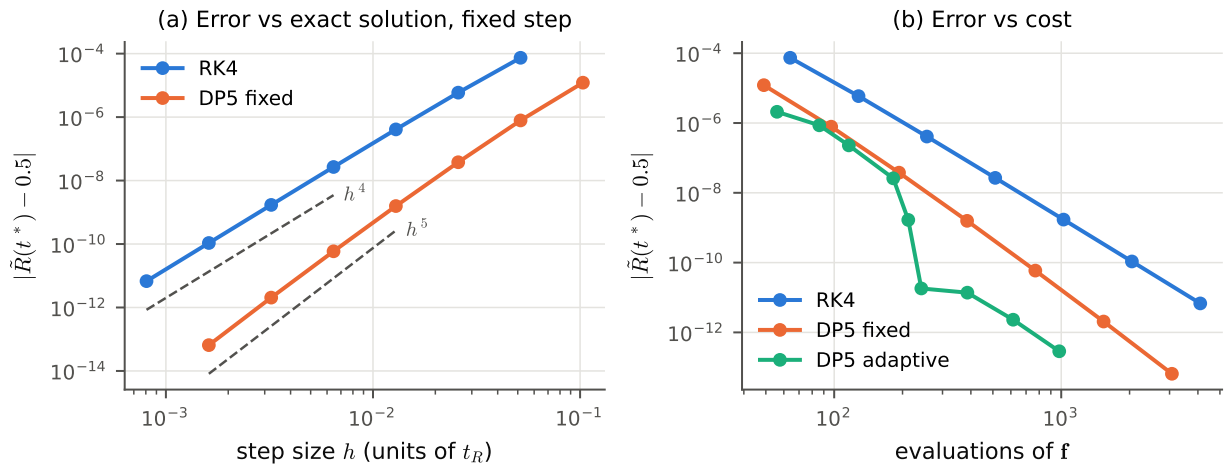


Figure 1: Case A (empty cavity) against the exact Rayleigh solution at $\tilde{R} = 0.5$. (a) Fixed-step error versus step size, with h^4 and h^5 reference slopes. (b) The same error versus cost, including adaptive DP5 at rtol $= 10^{-4}$ to 10^{-12}.

The rubric asks for two things: that the method converges at the expected rate, and that it converges to the solution of the right equations. A convergence study against a fine-step reference shows only the first, since a wrong term in $\mathbf{f}$ converges just as cleanly. We therefore check the answer against three things we know independently of our solver, one per test case (Section 3.1), before measuring rates on the full problem (Section 3.2).

3.1 Converging to the right answer

Case A: exact Rayleigh collapse. For the empty cavity, Eq. (5) gives the exact time to reach any radius. We evaluate its integral by Gauss–Legendre quadrature after the substitution $x = 1 - s^2$, which removes the endpoint singularity; this reproduces $\tilde{t}_c = 0.9146813565$ to 10^{-14} and is independent of any time stepping. Figure 1(a) integrates to the exact time $\tilde{t}^* = 0.825457$ at which $\tilde{R} = 0.5$ and plots the error in $\tilde{R}(\tilde{t}^*)$. Both fixed-step methods converge to the exact value, with observed orders 3.99 (RK4) and 4.97 (DP5) from the two finest steps. Deeper into the collapse the derivatives of $\mathbf{f}$ grow and the asymptotic range starts at smaller h, as Section 2.4 anticipates: at $\tilde{R} = 0.2$ the same step sizes give orders 3.90 and 4.48. Adaptive DP5 has no such difficulty. Run with rtol $= 10^{-12}$ until $\tilde{R} = 10^{-3}$, where $|\tilde{R}'| \approx 2.6 \times 10^4$, it reaches that radius at a time that differs from the exact one by 7×10^{-14}, and the state there satisfies the first integral Eq. (4) to a relative error of 1.2×10^{-11}.

Case B: energy conservation. With $\mu = 0$, the energy of Eq. (8) is exactly conserved, so its drift measures how well we solve the actual equations, gas and surface-tension terms included. For $\Pi = 10$ over $0 \le \tilde{t} \le 8$ (about four collapse and rebound cycles), the maximum relative drift falls from 2.6×10^{-3} to 4.6×10^{-10} as RK4 is refined from 400 to 12,800 steps, a factor of about $2^{4.3}$ per halving, and adaptive DP5 reaches 3×10^{-12} at rtol $= 10^{-12}$ (Fig. 3(b)).

Case C: linear response. At $\varepsilon = 10^{-4}$ the forced response is linear to $O(\varepsilon)$, so the steady amplitude from the full nonlinear equation must match the damped, driven oscillator of Eq. (10). For $R_0 = 10$ μm we integrate for 700 time units (about 14 damping times) and measure the

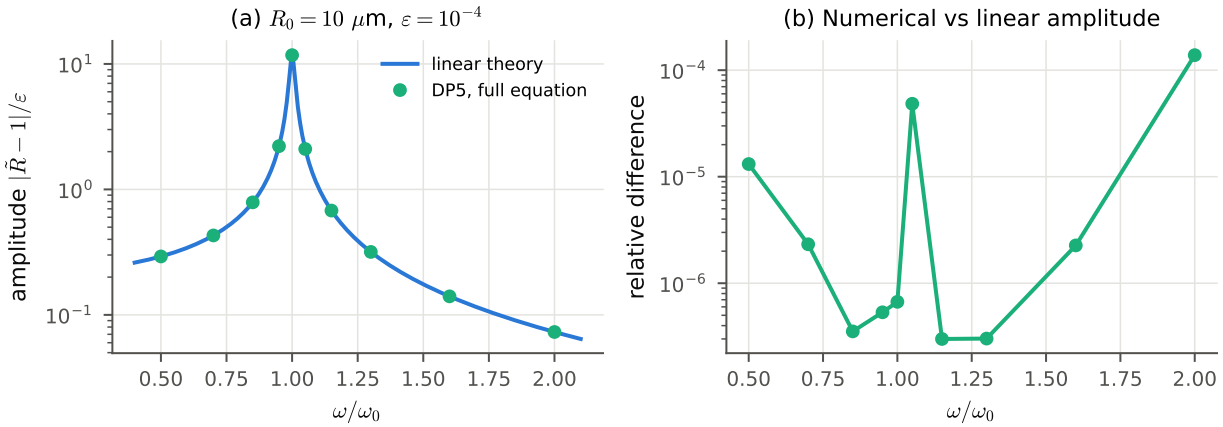


Figure 2: Case C at $\varepsilon = 10^{-4}$, $R_0 = 10$ μm. (a) Steady amplitude from the full equation (markers) and from linear theory, Eq. (10) (line). (b) Relative difference.

amplitude over the next 40 periods. Across $0.5 \leq \omega/\omega_0 \leq 2$, including the resonance peak where the amplitude is about $Q \approx 54$ times the static response, the numerical and linear amplitudes agree to better than 1.4×10^{-4} (Fig. 2), and to better than 5×10^{-5} at all but the highest frequency. The run also reproduces the values from the premise: $\omega_0 t_R = 2.161$, $f_0 = 342.5\,\mathrm{kHz}$ and $Q = 53.6$.

3.2 Convergence rate on the full problem

Figure 3(a) uses the baseline Case B ($R_0 = 100$ μm, $\Pi = 5$, all physics on) and measures the error in $\tilde{R}$ at $\tilde{t} = 3$, after the first collapse and into the rebound, against an adaptive DP5 reference at rtol $= 10^{-13}$. The observed orders are 3.96 for RK4 and 4.97 for fixed-step DP5, matching $p = 4$ and $p = 5$. The curves are irregular at the coarsest steps, where a step is comparable to the time spent near $R_{\min}$; this is the pre-asymptotic regime of Section 2.4.

3.3 Stability along collapse trajectories

Section 2.2 argued from the linearization about equilibrium that the problem is not stiff. Figure 4 checks this where it matters, during violent motion. At every accepted DP5 step (rtol $= 10^{-8}$) of a Case B collapse at $\Pi = 20$ and of 20 forcing periods of Case C at $\varepsilon = 0.3$, we compute the eigenvalues λ of the analytic Jacobian of Eq. (14) and plot $h\lambda$. The largest value is $|h\lambda| = 0.16$, against stability limits of 2.79 (RK4, real axis), 2.83 (RK4, imaginary axis), 3.31 (DP5, real axis) and 1.00 (DP5, imaginary axis). The step is therefore always set by accuracy. Panel (b) also shows how the controller works: $|h\lambda|$ stays near 0.1 throughout, so the step shrinks in proportion to the local time scale $1/|\lambda|$. Some eigenvalues have positive real part during the collapse. That is the physical growth of the collapsing motion, which an accurate method has to follow, not a numerical instability.

3.4 Choosing the simulation parameters

Q1 needs $R_{\min}$, the peak inward wall speed and the peak gas pressure. The last is the most sensitive, because $p_{g,\max} \propto R_{\min}^{-3\kappa}$ amplifies relative errors by about 4.2. Figure 5 measures all three against a reference at rtol $= 10^{-13}$ for the pressure ratios we sweep. To keep every quantity within 10^{-6} of the reference, far below what a plot can show, rtol $= 10^{-7}$ is enough for all Π. We use rtol $= 10^{-8}$

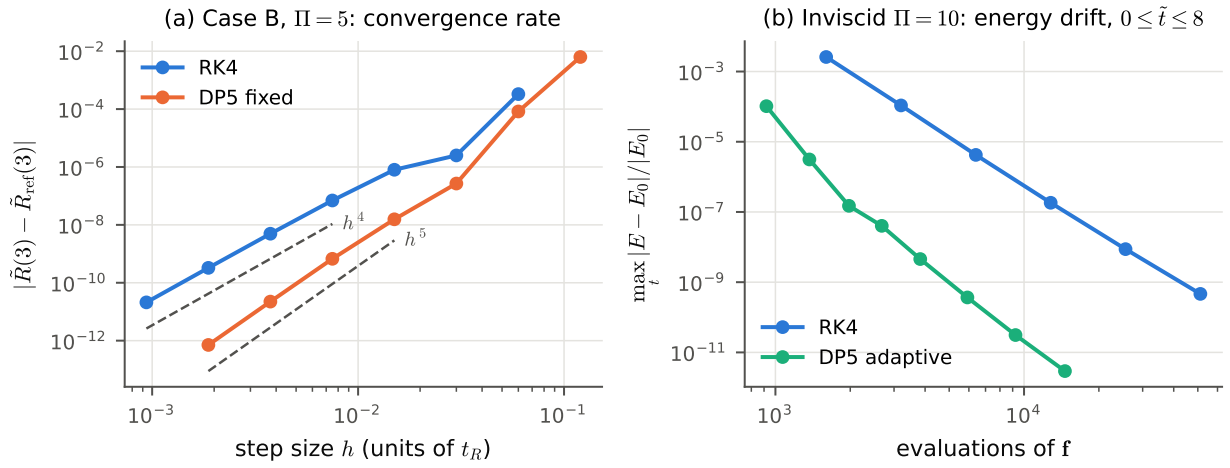


Figure 3: Case B. (a) Fixed-step error at $\tilde{t} = 3$ for $\Pi = 5$ with all physics on, against a tight-tolerance reference. (b) Maximum relative drift of the conserved energy, Eq. (8), for an inviscid collapse at $\Pi = 10$.

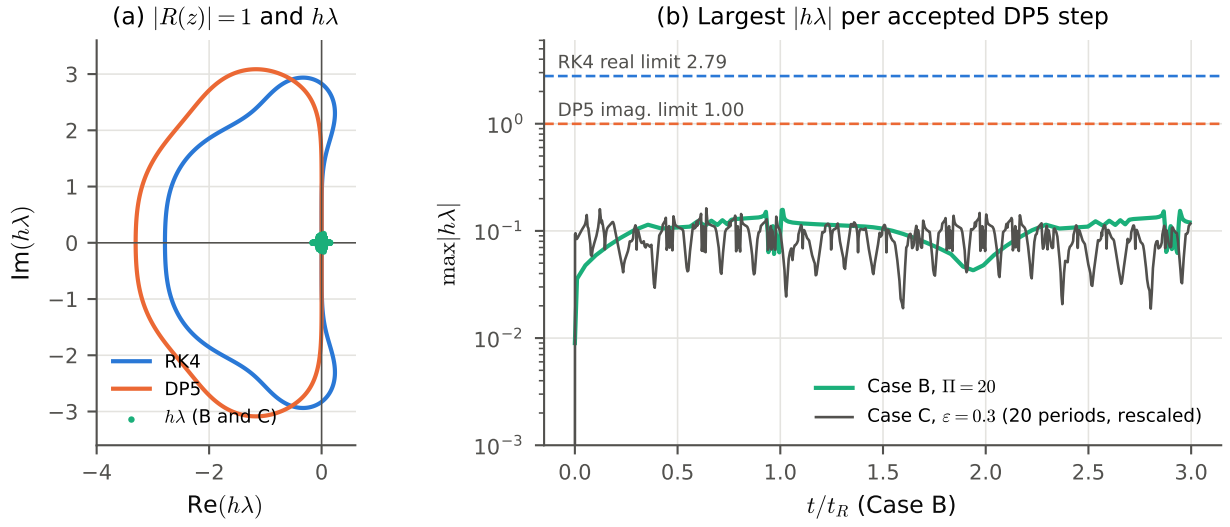


Figure 4: (a) Stability boundaries $|R(z)| = 1$ of RK4 and DP5 with the scaled eigenvalues $h\lambda$ from every accepted DP5 step of Case B ($\Pi = 20$) and Case C ($\varepsilon = 0.3$, $\omega = \omega_0$). (b) Largest $|h\lambda|$ per step against the tightest limit of each method. Case C time is rescaled to share the axis.

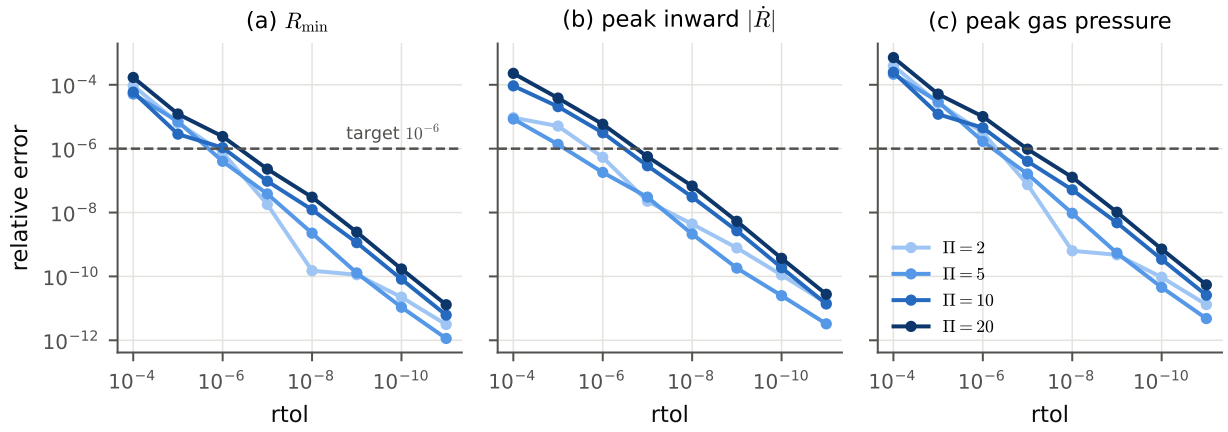


Figure 5: Relative error of the Q1 quantities versus DP5 tolerance for Case B at $R_0 = 100$ μm, measured against rtol $= 10^{-13}$. Dashed line: the 10^{-6} accuracy target.

and atol $= 10^{-10}$ for a margin of ten: the worst error is then 1.3×10^{-7} (peak gas pressure at $\Pi = 20$), and a run to the first collapse costs only 270 to 510 evaluations of $\mathbf{f}$. The extrema are located by the event search of Section 2.7, so they inherit this accuracy instead of being limited by where the steps happen to fall.

Adaptivity pays off most in the strongest collapses. For $\Pi = 20$ (Fig. 6) the accepted step varies by a factor of 166, from 6.8×10^{-2} to 4.1×10^{-4} (in units of t_R), shrinking sharply in the brief interval around each minimum. DP5 reaches an error of 9×10^{-9} at $\tilde{t} = 3$ with 1,508 evaluations (221 accepted and 30 rejected steps). Fixed-step RK4 needs about 4,750 steps, or 19,000 evaluations, to reach the same error reliably, so the adaptive method is about 13 times cheaper here. The same comparison gives factors of about 7 at $\Pi = 5$ and 8 at $\Pi = 10$, so the advantage grows with the strength of the collapse. Hence we use fixed-step RK4 only for the convergence studies above, and adaptive DP5 for every result in Section 4.

One finding matters for the results: at $\Pi = 20$ the peak wall Mach number is $|\dot{R}|_{\max}/c = 0.14$, above the cutoff of 0.1 set in Section 1.3. The upper end of our Π range is therefore outside the model's validity, and we mark it as such in Section 4.

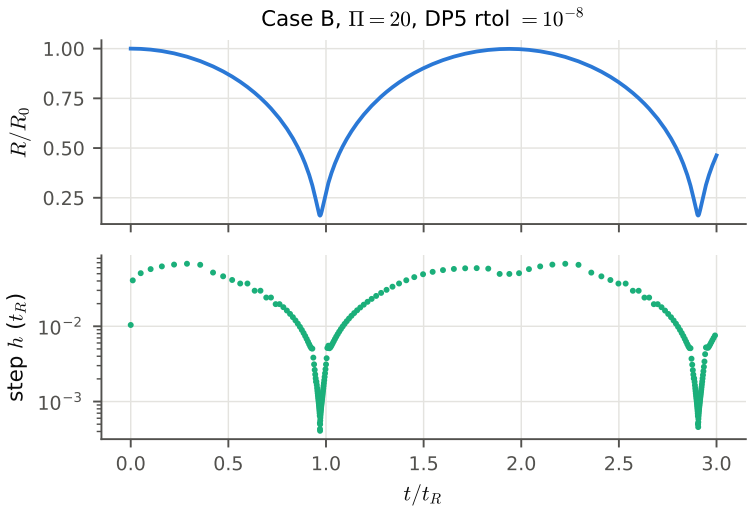


Figure 6: Case B at $\Pi = 20$: radius and accepted DP5 step size (rtol $= 10^{-8}$). The step collapses by two orders of magnitude around each minimum of R.

4 Results

We now use the Rayleigh–Plesset model to examine the three questions from Section 1.2. The pressure-step calculations use $\kappa = 1.4$ and the water properties in Table 1. We report the first collapse minimum and the most negative wall velocity reached before that minimum. The sinusoidal calculations use $R_0 = 10\,\mu$m; for each frequency, the amplitude is measured as half the peak-to-peak radius variation over the last 20 periods of a 200-period run, by which time the start-up transient has decayed. Every run uses adaptive DP5 with rtol $= 10^{-8}$ and atol $= 10^{-10}$ (Section 3.4); the forced runs also cap the step at 1/50 of a forcing period so that the radius maxima and minima are sampled. All results below are for the incompressible spherical-bubble model, so a predicted collapse speed above our chosen $0.1c$ limit must be interpreted cautiously.

4.1 Q1: Effect of pressure ratio and initial bubble radius

Figure 7 shows how the first collapse changes as the pressure ratio $\Pi = p_1/p_0$ increases. For every initial radius tested, the minimum radius decreases and the peak gas pressure rises sharply with Π. At $R_0 = 100\,\mu$m, increasing Π from 2 to 20 decreases $R_{\min}/R_0$ from about 0.704 to 0.160 and increases the peak gas pressure from about 0.438 to 222 MPa. The latter change is much larger because the polytropic relation $p_g = p_{g0}(R_0/R)^{3\kappa}$ amplifies a reduction in radius.

Π	$R_{\min}/R_0$	$\lvert\dot{R}\rvert_{\max}$ (m/s)	$p_{g,\max}$ (MPa)	$\lvert\dot{R}\rvert_{\max}/c$
2	0.7043	5.29	0.438	0.0036
5	0.4156	25.36	4.014	0.0171
10	0.2636	72.18	27.16	0.0487
20	0.1599	211.71	221.56	0.1429

Table 3: First-collapse quantities for Case B at $R_0 = 100\,\mu$m, $\kappa = 1.4$.

Bubble size also affects the collapse, but less strongly across the 10–1000 μm range than the pressure ratio does. For example, at $\Pi = 5$, the computed minimum-radius ratios are 0.4426,

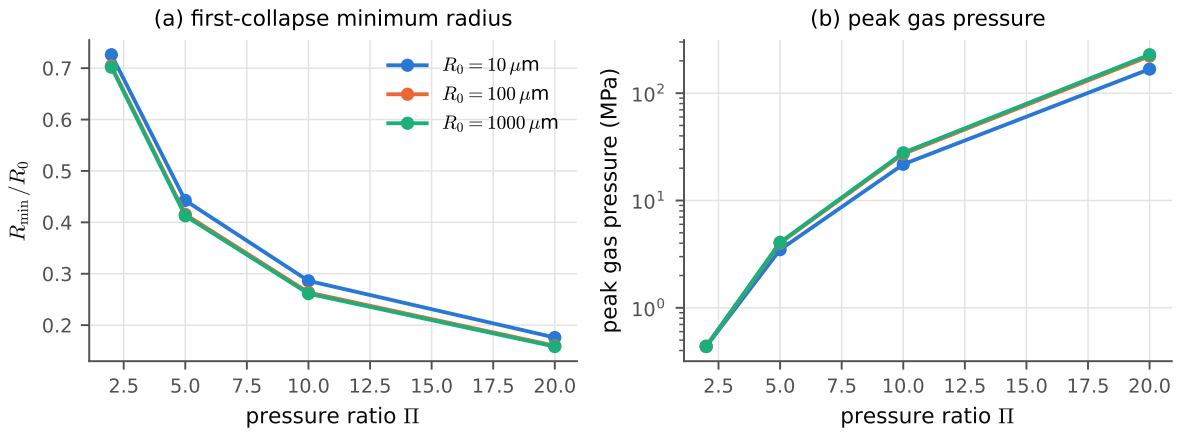


Figure 7: First-collapse minimum radius and peak gas pressure versus the pressure ratio Π for three initial bubble radii. The gas pressure is evaluated from the minimum radius using the polytropic model.

0.4156, and 0.4127 for $R_0 = 10$, 100, and 1000 μm, respectively. The curves approach one another for larger bubbles, consistent with the dimensionless surface-tension and viscous terms becoming less important as R_0 increases. The gas pressure nevertheless remains crucial for stopping collapse at a finite radius; our results do not identify a universal pressure ratio beyond which gas compression can be neglected at the minimum.

The validity check is important at high pressure ratios. For $R_0 = 100\,\mu$m and $\Pi = 20$, the predicted peak wall Mach number is about 0.143, above the 0.1 cutoff adopted in Section 1.3. Even though the numerical integration resolves this solution, liquid compressibility may substantially affect the actual peak pressure and wall speed. We therefore treat this point as a trend predicted by the model rather than a quantitatively reliable physical value.

4.2 Q2: Response to oscillating pressure

Figure 8 compares the steady oscillation amplitude over a coarse forcing-frequency sweep at three pressure amplitudes ε. For the smallest forcing amplitude, $\varepsilon = 0.01$, the response has a narrow maximum near $\omega/\omega_0 = 1$, as expected for a weakly damped oscillator. The half peak-to-peak radius variation is about $0.106R_0$ at this sampled frequency, compared with about $0.0115R_0$ at $\omega/\omega_0 = 0.9$. This confirms that forcing near the linear natural frequency produces a much larger response than off-resonance forcing.

As the forcing increases, the response becomes nonlinear and asymmetric about the initial radius. At $\varepsilon = 0.3$ and $\omega/\omega_0 = 0.9$, the simulated bubble reaches approximately $R_{\max}/R_0 = 1.766$ and $R_{\min}/R_0 = 0.465$, whereas the response at $\omega/\omega_0 = 1$ is smaller by the same amplitude measure. This indicates a shift in the strongest sampled response toward a frequency below the small-amplitude natural frequency. However, the current grid has a spacing of $0.1\omega_0$ and is not fine enough to locate the peak precisely. A narrower frequency sweep around the observed maximum would be needed to quote the size of the shift.

4.3 Q3: Roles of surface tension and viscosity

To isolate the forces resisting collapse, we compare the full model with calculations in which viscosity or surface tension is removed while retaining the appropriate initial equilibrium gas pressure.

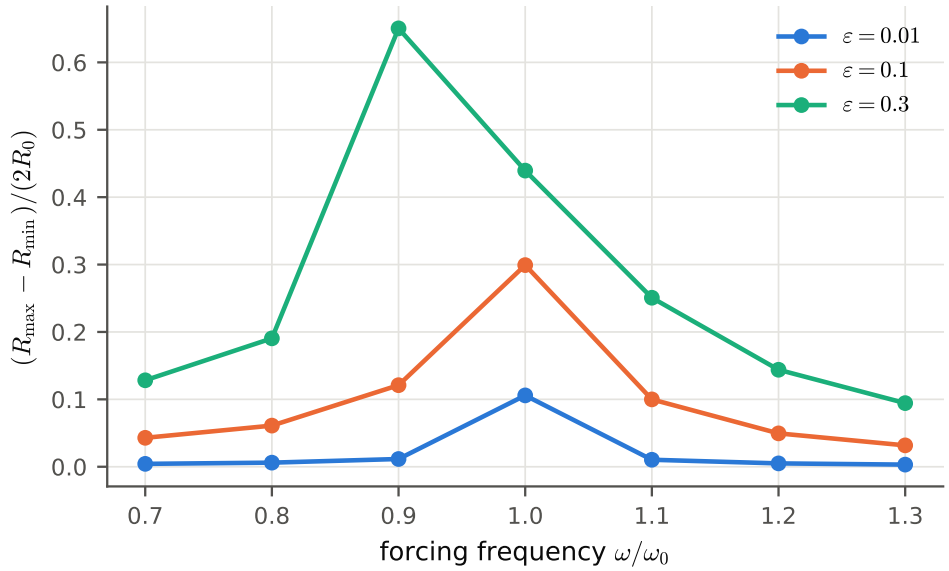


Figure 8: Half peak-to-peak normalized radius response for Case C at $R_0 = 10\,\mu$m and three forcing amplitudes. Points are sampled frequencies, not precisely located resonance peaks.

Figure 9 uses $\Pi = 5$ and shows the first minimum radius for initial sizes from 1 to 100 μm.

At $R_0 = 100\,\mu$m the minimum-radius ratio changes only from 0.4156 to 0.4152 when viscosity is removed and to 0.4128 when surface tension is removed. At $R_0 = 1\,\mu$m, these differences are much greater: the full model predicts 0.6135, removing viscosity gives 0.5881, and removing surface tension gives 0.4533. In the smaller bubble, both effects materially resist collapse, with surface tension having the larger influence under these particular conditions. The result is consistent with the dimensionless groups of Section 1.5, in which surface-tension and viscous contributions become larger as the initial radius decreases.

4.4 Overall interpretation and limitations

The simulations point to three conclusions. First, pressure ratio strongly controls the severity of the first collapse, and the polytropic gas law turns modest changes in minimum radius into much larger changes in peak pressure. Second, forcing close to the natural frequency produces a strong response, but the largest-response frequency can shift once the pressure oscillation is no longer small. Third, surface tension and viscosity affect micron-scale bubbles more strongly than larger ones, so changing the initial radius changes how strongly each term limits the collapse.

These are conclusions within the spherical, incompressible Rayleigh–Plesset model. As discussed in Section 1, it does not resolve shock waves, nonspherical collapse, or the thermal physics of cryogenic propellants. Furthermore, frequency maxima located from a coarse sweep should not be presented as exact resonance frequencies. Taken together with the convergence and stability checks of Section 3, the results demonstrate what the implemented numerical method can resolve while making clear where the underlying physical model, rather than numerical error, becomes the limiting factor.

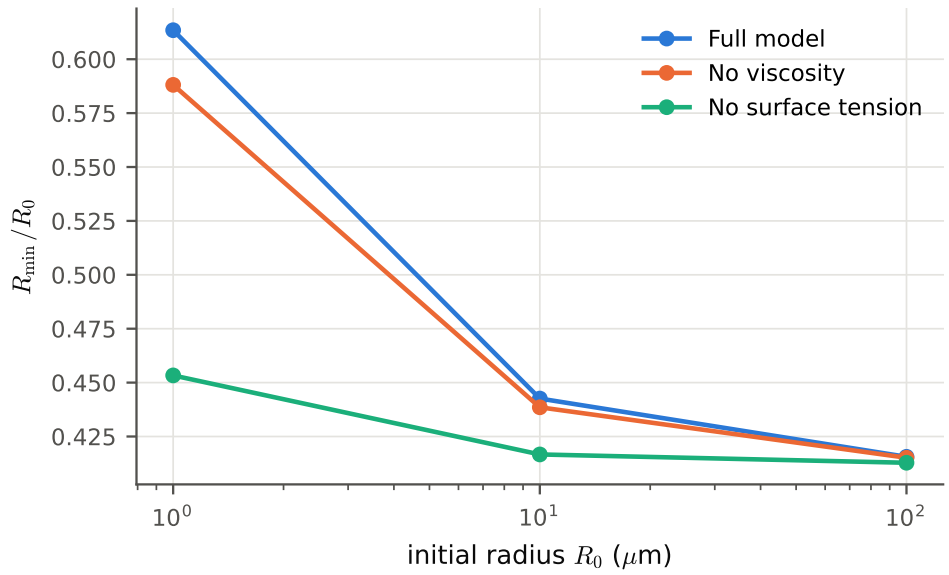


Figure 9: Effect of removing viscosity or surface tension on the first minimum radius in Case B at $\Pi = 5$, $\kappa = 1.4$. The gas pressure is adjusted for the corresponding initial equilibrium condition.

5 Contributions

Member	Rubric items owned (1 to 4)	Other duties	Est. %
Naveen Koshy	1 Premise, 3 Implementation	GitHub repository, report build	34
Ava Hoppe	2 Method	–	33
Alison Nazaruk	4 Results	–	33

6 Reproducibility

All code is available at `https://github.com/nkoshy2/rayleigh-plesset-ae370`. The integrators live in `src/rp/`, each report figure is produced by one script in `experiments/`, and the tests in `tests/` can be run with `pytest`. No library time integrator is used to advance the solution. After `pip install -r requirements.txt` and `pip install -e .`, the command `pytest` runs the 24 unit tests and `python experiments/run_all.py` regenerates every figure in this report, printing the numbers quoted in the text.

Acknowledgments

We used generative AI, mainly Claude (Anthropic, through Claude Code), at most stages of the project. We are responsible for the final content of this report and the code.

Ideation and premise. Claude set up the repository skeleton and drafted the premise for our topic: the motivation, research questions, initial value problem, parameter table, test cases and nondimensionalization. It also checked the water properties against the NIST WebBook and the references and the Mach-number claims with web searches; from that check we updated the property

values and restated the 0.1 wall-Mach-number cutoff as our own choice, since no source gives a single accepted value.

Method selection and justification. We wrote the method section. Claude reviewed it against the rubric and corrected errors in the cross-references, the wording of the order conditions, the damping order of RK4, the imaginary-axis stability limit of the Dormand–Prince pair and the description of event location, and added missing citations.

Implementation and testing. Claude wrote the model, the RK4 and adaptive Dormand–Prince integrators (no library time integrator is called), the unit tests and the verification experiments, and drafted the implementation section.

Figures and results. We ran the parameter sweeps for Q1–Q3 and wrote the results section. In a final review, Claude checked the report against the project rubric, added scripts to the repository that regenerate the three results figures, reran every experiment to confirm that the numbers quoted in Sections 3 and 4 match the code's output, and flagged statements that went beyond what we had computed, which we then removed or reworded.

Writing and tooling. Claude also helped install and debug the LaTeX build, create the GitHub repository, and draft this acknowledgment from our AI usage log (`docs/ai-usage-log.md` in the repository).

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